SCHEMATICS

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVE
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

TITLE PAGE

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER		REV A
	APPROVED DATE 20/10/2026	SCALENTC		SHEET 001 OF 015	

INDEX

SL. NO.	DRAWING NO.	DRAWING DESCRIPTION
1	SH001	TITLE PAGE
2	SH002	INDEX SHEET
3	SH003	HOW TO READ SCHEMATIC DIAGRAM
4	SH004	HOW TO READ SCHEMATIC DIAGRAM
5	SH005	LEGENDS
6	SH006	GENERAL ARRANGEMENT (GA)
7	SH007	PANEL LAYOUT
8	SH008	POWER DISTRIBUTION CIRCUIT
9	SH009	CONTROL POWER
10	SH010	IPC AND HMI
11	SH011	FIELD CONNECTIONS
12	SH012	FIELD CONNECTIONS
13	SH013	FIELD CONNECTIONS
14	SH014	BILL OF MATERIALS (BOM)
15	SH015	CABLE HARNESS

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

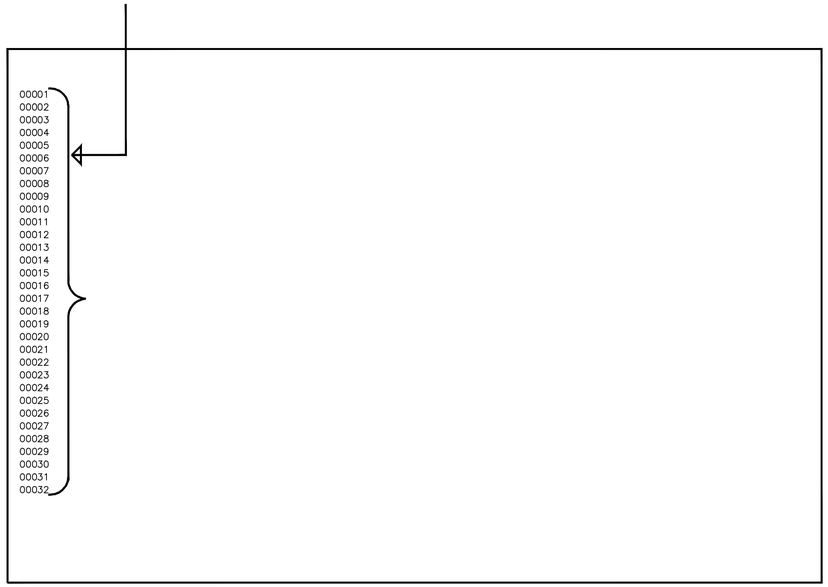
INDEX SHEET

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
APPROVED	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 002 OF 015	

Schematic Border / Template

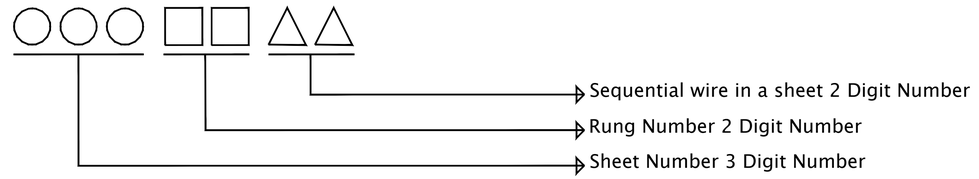
I) Each Border/Template has 32 rungs or rows.
II) Rung numbers are used in wire numbering, Component Tag/Device ID, Cross referencing.
III) First 3 digits: Sheet number padded to three digits
IV) Last 2 digits: Rung number counting sequentially from 01 to 32.

000 Sheet numbers , Rung numbers – 01 to 32



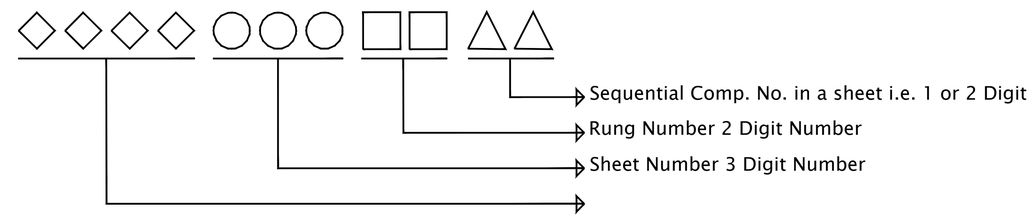
Determination of Wire Number & Device ID/Component Tag

I) Wire Number Format (7 Digit number)



Example :- 0021101, 0081009, 1050112, etc.

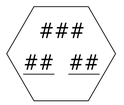
II) Device ID / Component Tag format



Example :- FL007081, GBLK0071225, BSBR0080101, etc.

III) Cross reference format

Page number
Page and rung number



Rung number

WIRE – Wire number
DESC – Description

REVISION HISTORY						
REV	DESCRIPTION	DATE	APPROVED			
REV.A	RELEASED FOR PRODUCTION	20/10/2026				
REV.0	INITIAL RELEASE	10/10/2026				

DRIVE TRAIN ADAPTER PANEL

HOW TO READ SCHEMATIC DIAGRAM

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D / DRAWING NUMBER	REV A
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 003 OF 015

Wire Color Abbreviation

COLOR NAME	ABBREVIATION
Black	BK
Blue	BL
Brown	BN
Green	GN
Gray	GY
Orange	OR
Pink	PK
Purple	PR
Red	RD
Shield	SHLD
Slate	SLT
Tan	TN
Turquoise	TQ
Violet	VT
White	WH
Yellow	YL
Black Green	BKGN
Black Red	BKRD
Blue Black	BLBK
Blue White	BLWH
Brown Green	BNGN
Brown White	BNWH
Green Yellow	GNYE
Gray Pink	GYPK
Gray White	GYWH
Orange Black	ORBK
Orange Blue	ORBL
Orange Red	ORRD
Orange White	ORWH
Red Black	RDBK
Red Blue	RDBL
Red White	RDWH
White Black	WHBK
White Blue	WHBL
White Brown	WHBN
White Green	WHGN
White Orange	WHOR
White Red	WHRD
White Yellow	WHYL
Yellow Brown	YLBN
Yellow White	YLWH

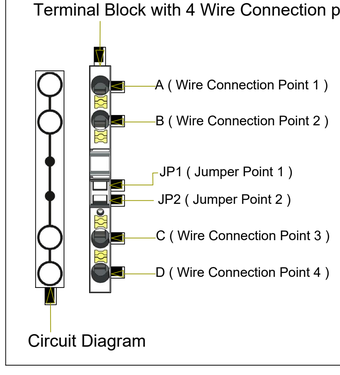
Wire Gauge Conversion Table

Component Code

COMPONENT	CODE
Adaptor	ADAP
Audible Buzzer	AB
Audible Horn	AH
Audible Alarm Indicator	IND
Battery	BAT
Busbar	BSBR
Cable / Multicore Cable	CBL
Camera	CAM
Capacitor	CP
Charger	CHG
Circuit Breaker	CB
Connector Plug	CONP
Connector Receptacle	CONR
Compressor	CMP
Contactor	CR
Current Transformer	CT
Disconnector Switch	DS
Diode	D
Device (Generic)	DEV
Fan	FN
Ferrule	FRL
Filter	FL
Fuse	FU
Fuse Holder	FH
Fuse Block	FB
Ground	GND
Heater	HTR
Lamp & LED	LT
Limit Switch	LS
Logic Module	LM
Motor	MTR
Network Switch	NW
Overload Relay	OLR
Pendant	PDT
Power Supply Unit	PSU
Pressure Switch	PS
Printed Circuit Board	PCB
Programmable Logic Controller	PLC
Push Button	PB
Pump	PUMP
Control Relay / Relay	RLY
Resistor	R
Ring Terminal	RING
Robot	RBT
Scanner	SCN
Selector Switch	SS
Sensor	SNSR
Server	SER
Solenoid	SOL
Splice	SP
Switch	SW
Terminal Block	TB
Thermocouple	TC
Time-Delay Relay	TR
Transformer	TRF

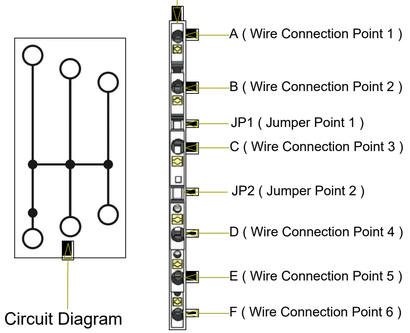
Terminal Block & its Accessories

Terminal Block with 4 Wire Connection points & 2 Jumper Points



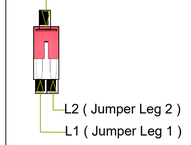
Circuit Diagram

Terminal Block with 6 Wire Connection points & 2 Jumper Points

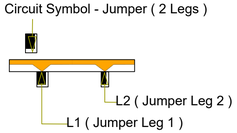


Circuit Diagram

Jumper (2 Legs)



Circuit Symbol - Jumper (2 Legs)



REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRAWN VIJAYAKUMAR V S | DRAWN DATE 10/10/2026 | SIZE D | DRAWING NUMBER | REV A

APPROVED | APPROVED DATE 20/10/2026

SCALE NTC | SHEET 004 OF 015

DRIVE TRAIN ADAPTER PANEL

HOW TO READ SCHEMATIC DIAGRAM

	2 Pole Miniature Circuit Breaker
	3 Pole Miniature Circuit Breaker
	Circuit Breaker 3 pole
	Fuse
	Circuit Breaker with Magnetic & Thermal Protection
	Coil (Relay, Contactor)
	Auxiliary Contact Normally Open/1SPST-NO (Relay, Contactor)
	Auxiliary Contact Normally Close/1SPST-NC Relay, Contactor)
	Power Supply Unit (AC-DC Converter)
	Battery / Cell

	Illuminated light
	Push button NO
	Push button NC
	Mushroom Push Button NC

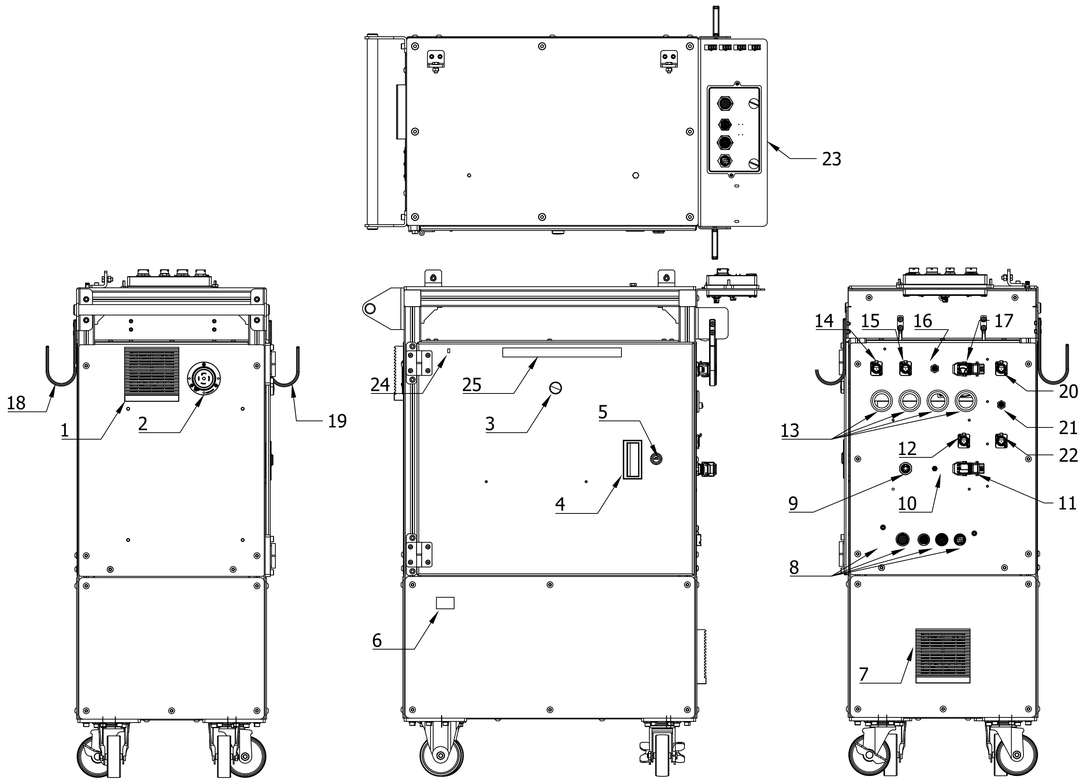
REVISION HISTORY

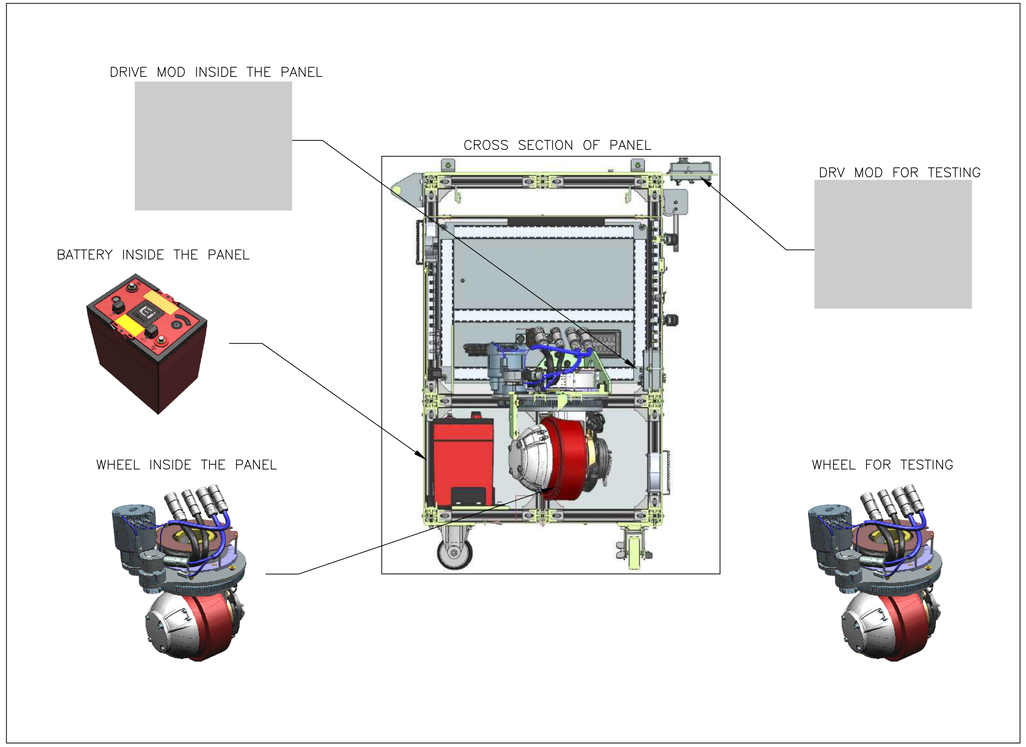
REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

LEGENDS

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 005 OF 015	





IDENTIFICATION		
ITEM	TAG / LABEL	DESCRIPTION
1	EXHAUST FAN	Panel Enclosure Exhaust Cooling Fan
2	208V, 3P POWER INPUT FROM COMMON CONTROL PANEL	3-Phase Main Power Input Connection from COMMON CONTROL PANEL
3	208V AC ON	Main Power Status Indicator LED / Light
4	GRAB HANDLE	Panel Door Grab Handle
5	LOCK	Enclosure Access Door Key Lock
6	BATTERY STATUS	Battery Charge Level / State Indicator
7	INLET FAN	Panel Enclosure Intake Cooling Fan
8	WHEEL TESTING CABLES	Interface Cables for Wheel Testing

9	DRIVE TRAIN 24V DC $\frac{7}{8}$"	24V DC Power Interface Connection for Drivetrain via 7/8-inch Connector
10	DRIVE TRAIN 24V DC M8	24V DC Power Interface Connection for Drivetrain via M8 Connector
11	DRIVE TRAIN 48V DC	48V DC High-Power Supply Connection for Drivetrain Test
12	DRIVE TRAIN ECAT IN	EtherCAT Industrial Ethernet Communication Input for Drivetrain Test
13	DRIVE MOD TEST CABLES	Interface Cable for Drive Module Testing
14	DRIVE ECAT IN	EtherCAT Industrial Ethernet Communication Input for Drive Module Testing
15	DRIVE ECAT OUT	EtherCAT Industrial Ethernet Communication Output for Drive Module Testing

16	DRIVE 24V DC	24V DC Control Power Supply for Drive Module Testing
17	DRIVE 48V DC	48V DC Primary Power Supply for Drive Module Testing
18	J HOOK	Cable Support J-Hook for Drive test harness cables
19	J HOOK	Cable Support J-Hook for Wheel test harness cables
20	IPC COMMUNICATION IN	Coupler communication
21	E STOP OUT	Emergency Stop (E-Stop) Safety Circuit Loop Output Connection
22	RJ45 SPARE	Unassigned RJ45 Ethernet Communication Port
23	TEST DRIVE HOLDER	Mechanical Mounting Bracket / Fixture for Secure Test Drive Module
24	LS2	Door Limit switch
25	LAMP2	For illumination

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRAWN VIJAYAKUMAR V S

DRAWN DATE 10/10/2026

APPROVED DATE 20/10/2026

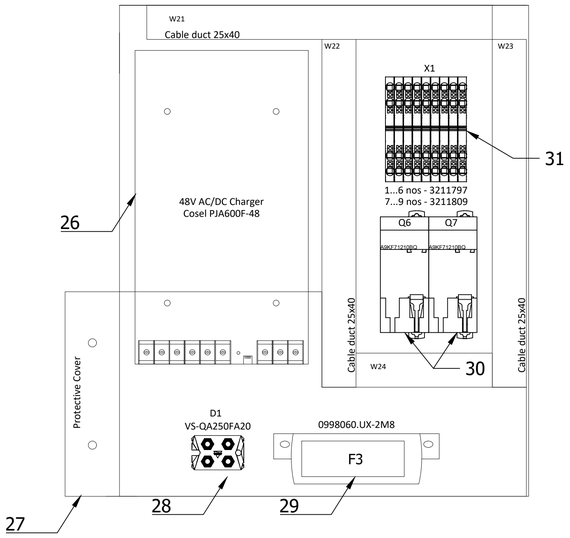
DRIVE TRAIN ADAPTER PANEL

GENERAL ARRANGEMENT (GA)

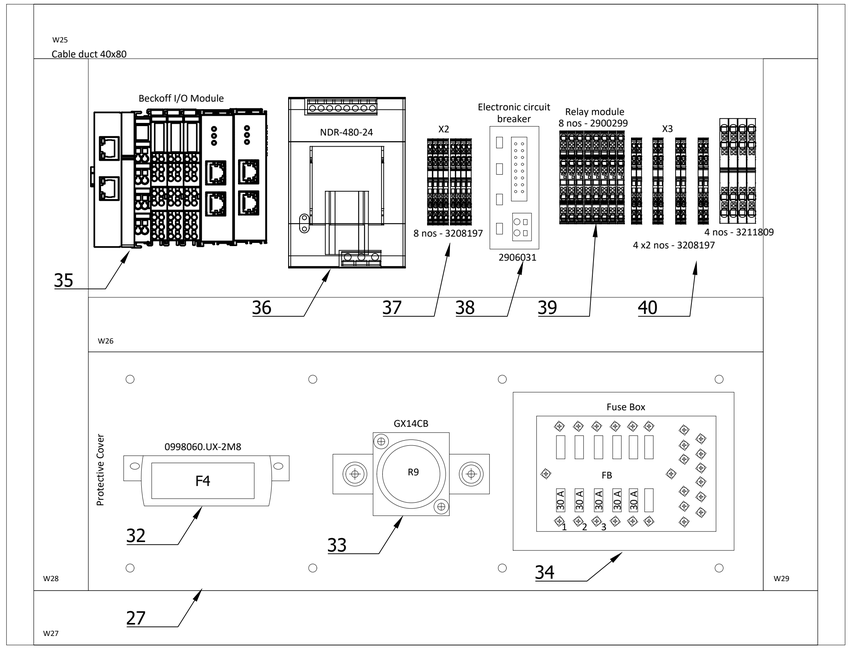
SIZE D | DRAWING NUMBER | REV A

SCALE NTC | SHEET 006 OF 015

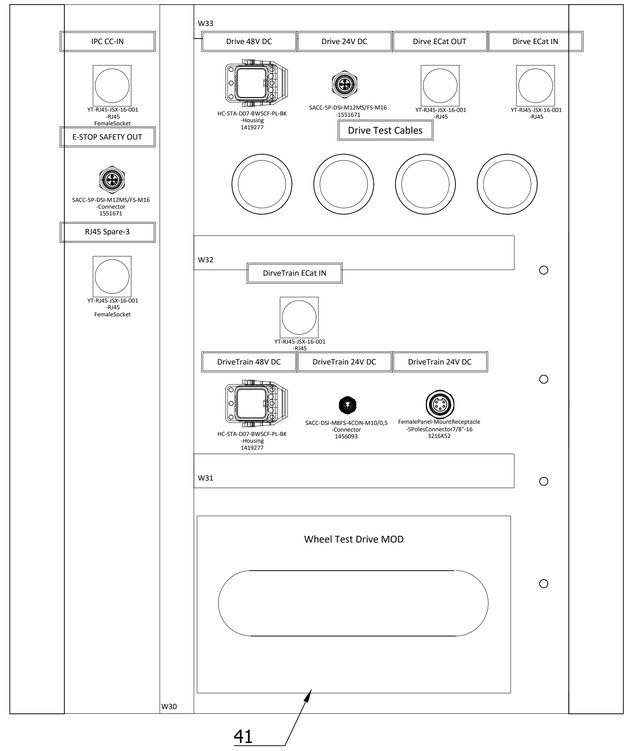
LEFT SIDE



PANEL BASE PLATE



RIGHT SIDE



IDENTIFICATION		
ITEM	TAG / LABEL	DESCRIPTION
26	48V AC/DC Charger	Cosel PJA600F-48, Power Supply, 48 V DC
27	Protective Cover	Acrylic protective cover, transparent shield
28	D1	VS-QA250FA20 Diodes & Rectifiers 200V 250A Module
29	F3	02981028HXFC with 0998060.UX-2M8
30	Q6,Q7	10A, 2P, MCB
31	X1	1...6 nos - 3211797 7...9 nos - 3211809
32	F4	02981028HXFC with 0998060.UX-2M8
33	R9	Contactors - Electromechanical CONTACTOR
34	FB	Fuse Block 30 A 80V 12 Circuit Blade Chassis Mount

35	Beckoff I/O Module	01 : EK1101-0010 — EtherCAT Coupler 02 : EL1904 — Safety Digital Input 03 : EL6910 — TwinSAFE Logic 04 : EL2409 — Digital Output 05 : EK1122 — EtherCAT Junction 06 : EK1122 — EtherCAT Junction 07 : EL9011 — Bus End Terminal
36	PSU	DIN Rail Power Supply, 24 V DC, 20 A, 480 W
37	X2	8 nos - 3208197
38	CB	CBMC E4 24DC/1-4A NO - Electronic circuit breaker
39	R1-8	8Nos. PLC-RPT- 24DC/21 - Relay module
40	X3	4 x2 nos - 3208197 4 nos - 3211809
41	Wheel Test Drive MOD	DRIVE MODULE, GEN2

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

PANEL LAYOUT

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 007 OF 015	

208V, 3P POWER OUTPUT TO DRIVETRAIN TESTER CONP008011

208V, 3P POWER INPUT FROM COMMON CONTROL PANEL CONR008012

X
Y
Z
PE

008011 10AWG_RED
008021 10AWG_BLK
008031 10AWG_BLU
008041 10AWG_GRN-YLW

DOOR LIMIT SWITCH LS2 LS008011

008012 12AWG_RED
008013 12AWG_RED
008022 12AWG_BLK

PANEL LAMP LAMP 2 LT008011

LT008012 208V AC ON
008014 12AWG_RED
008015 12AWG_BLK

FN008031 INLET FAN
008033 12AWG_RED
008034 12AWG_BLK

FN008051 EXHAUST FAN
008051 12AWG_RED
008052 12AWG_BLK

008032

Q6 10A, 2P, MCB CBO008101
008101 12AWG_RED
008102 12AWG_RED
008111 12AWG_BLK
008112 12AWG_BLK

PSU PSU008101
L
PE
N
V+
V−
20AWG_BLU
008103
008113
20AWG_BLU

008103 009 009-01 24V DC
008113 009 009-02 0V DC

008121 14AWG_GRN-YLW

Q7 10A, 2P, MCB CBO008141
008141 12AWG_RED
008142 12AWG_RED
008151 12AWG_BLU
008152 12AWG_BLU

48V DC CHARGER PSU008141
L
PE
N
V+
V−
008143 10AWG_RED

008181 009,011,011,012,012,013,013,013,013,013 009-19,011-01,011-16,012-01,012-16,08,08,08,08,21 GND
X3 9 008181 14AWG_GRN-YLW

008211 011 0X1-03 48V DC FROM BATTERY
008211 10AWG_RED +1 FB FU008211 30A F5

008231 012 0X2-03 48V DC FROM BATTERY
008231 10AWG_RED +2 FB FU008233 30A F6

008251 013 0X3-06 48V DC FROM BATTERY
008251 10AWG_RED +3 FB FU008251 30A F7

008271 011 0X1-02 0V DC FROM BATTERY
008271 10AWG_BLK −1 FB

008291 012 0X2-02 0V DC FROM BATTERY
008291 10AWG_BLK −2 FB

008311 013 0X3-07 0V DC FROM BATTERY
008311 10AWG_BLK −3 FB

+ FB
− FB

009042 009 009-04 24V FOR R9
009042 20AWG_BLU
R9 48V SUPLLY
NO 008-23 NC
009021 20AWG_BLU
009021 009 009-09 0V

008232 10AWG_RED
R9 008-21 48V SUPLLY
008233 10AWG_RED
FU008232 60A F4
008234 10AWG_RED
IN−LINE, FREE−HANGING FUSE HOLDER FU008231 20A F5
008235 10AWG_RED
D1 D008231

008234 4AWG_BLK
BAT008261 BATTERY INSIDE THE PANEL
008292 4AWG_BLK
008292 10AWG_BLK
FB -

GENERAL NOTES:
• DISCONNECT ALL POWER SOURCES BEFORE REMOVING COVERS.
WIRING & CONNECTION SPECIFICATIONS:
• CONNECT +VE TERMINAL BATTERY WIRE TO IN TERMINAL OF F4.
• CONNECT -VE TERMINAL BATTERY WIRE TO FB FUSE BOX -VE.
BATTERY BYPASS / ISOLATION INSTRUCTIONS:
• IF BATTERY IS NOT USED, THE TWO (008234,008292) 4AWG CABLES MUST NOT HANG FREE.
• TO SAFELY OPERATE THE MACHINE WITHOUT A BATTERY, BOTH 4AWG CABLES MUST BE PROPERLY INSULATED AND ISOLATED.

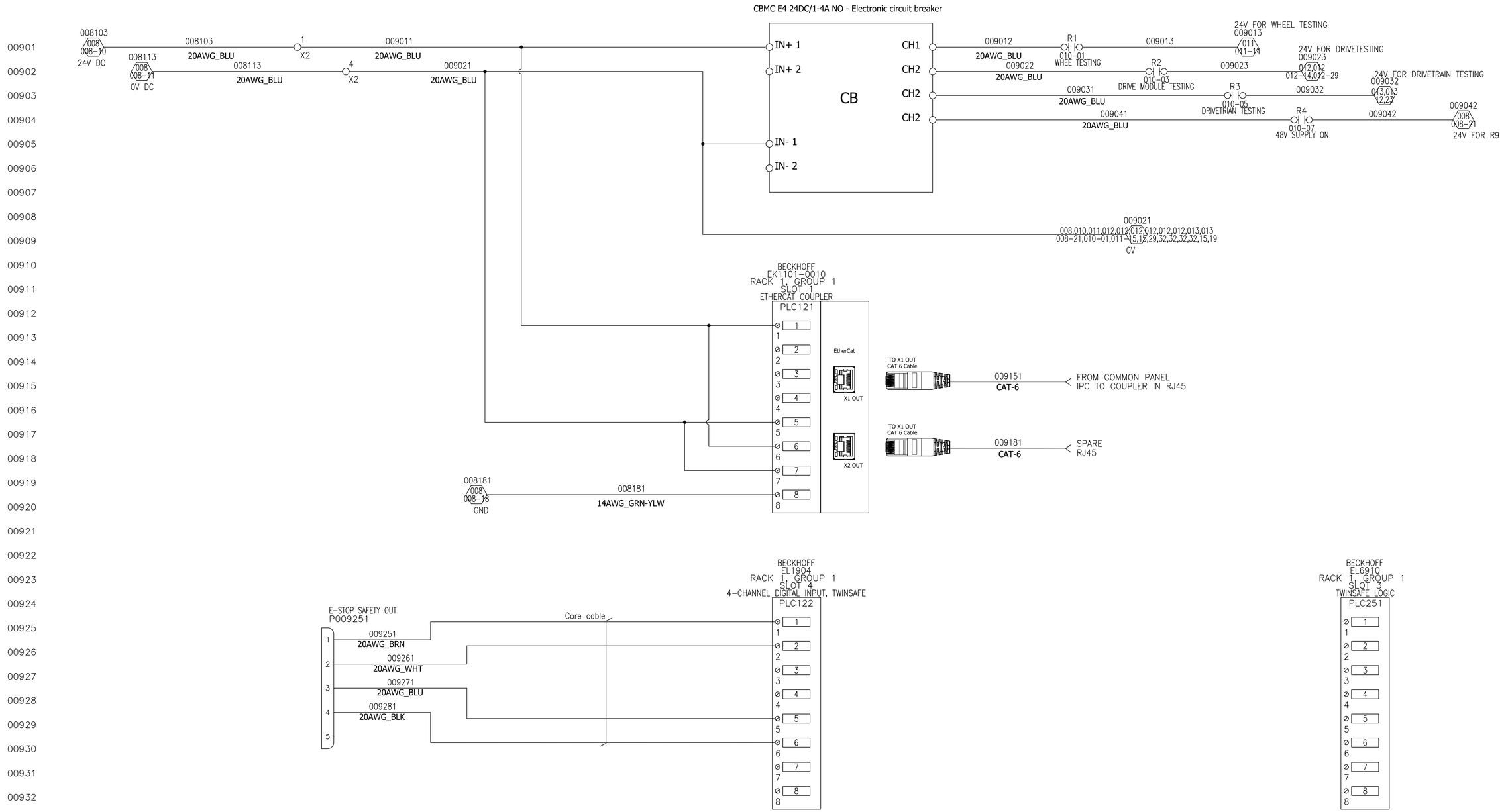
REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL
POWER DISTRIBUTION CIRCUIT

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER
APPROVED	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 008 OF 015

REV A

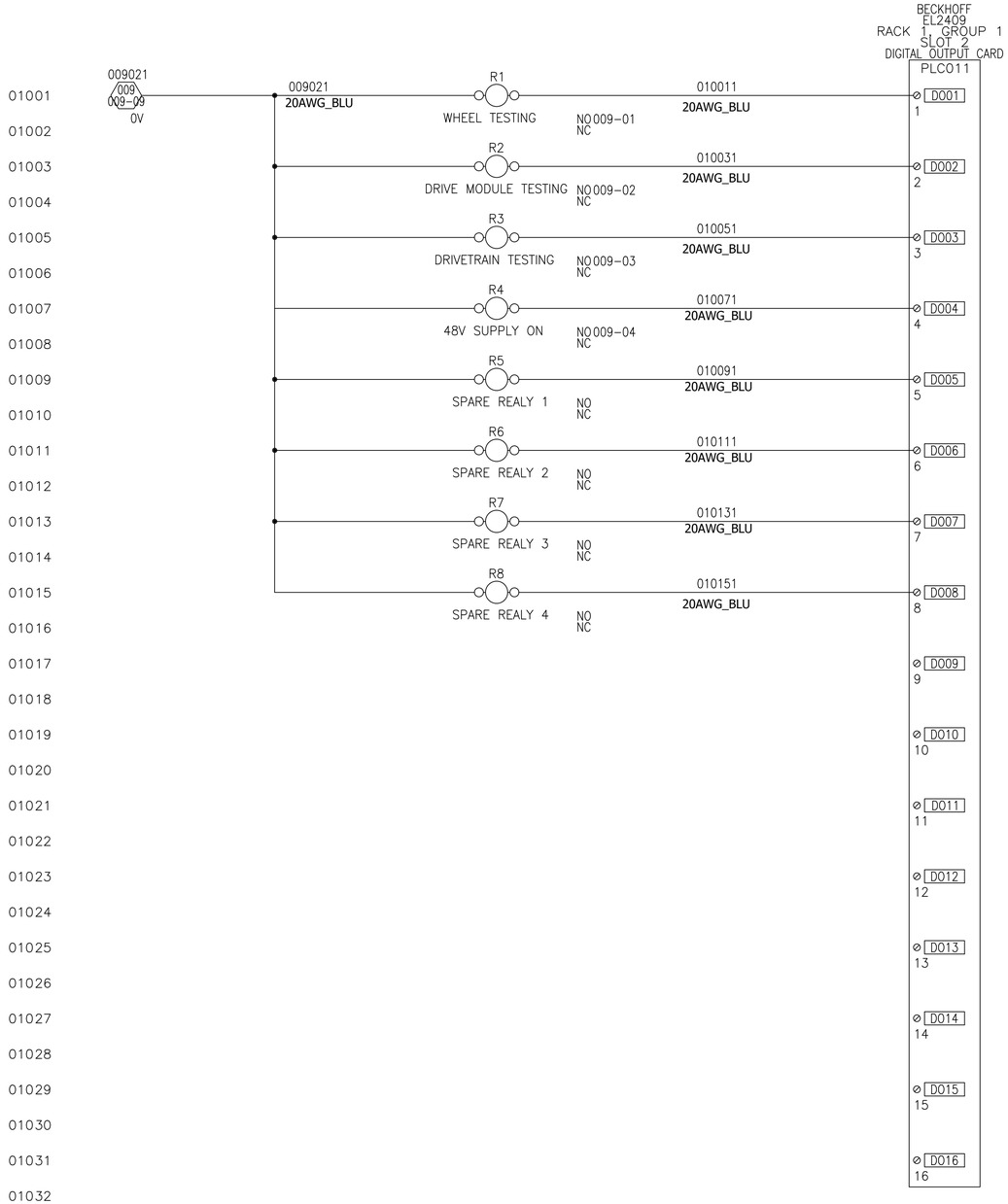


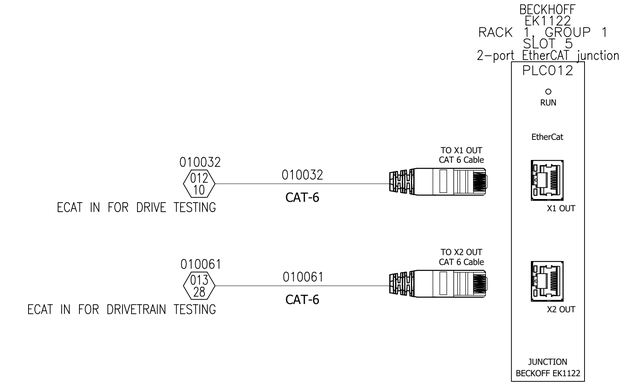
REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

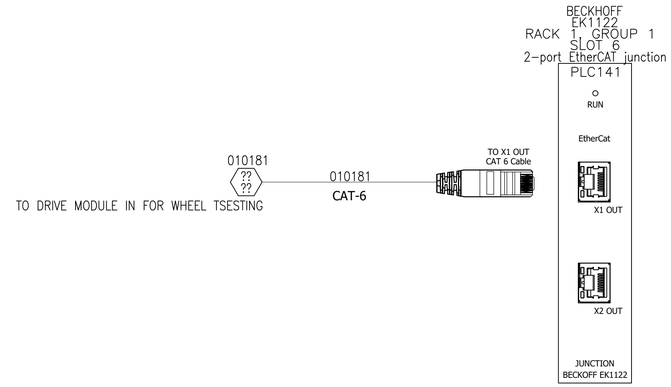
DRIVE TRAIN ADAPTER PANEL

CONTROL POWER

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 009 OF 015	



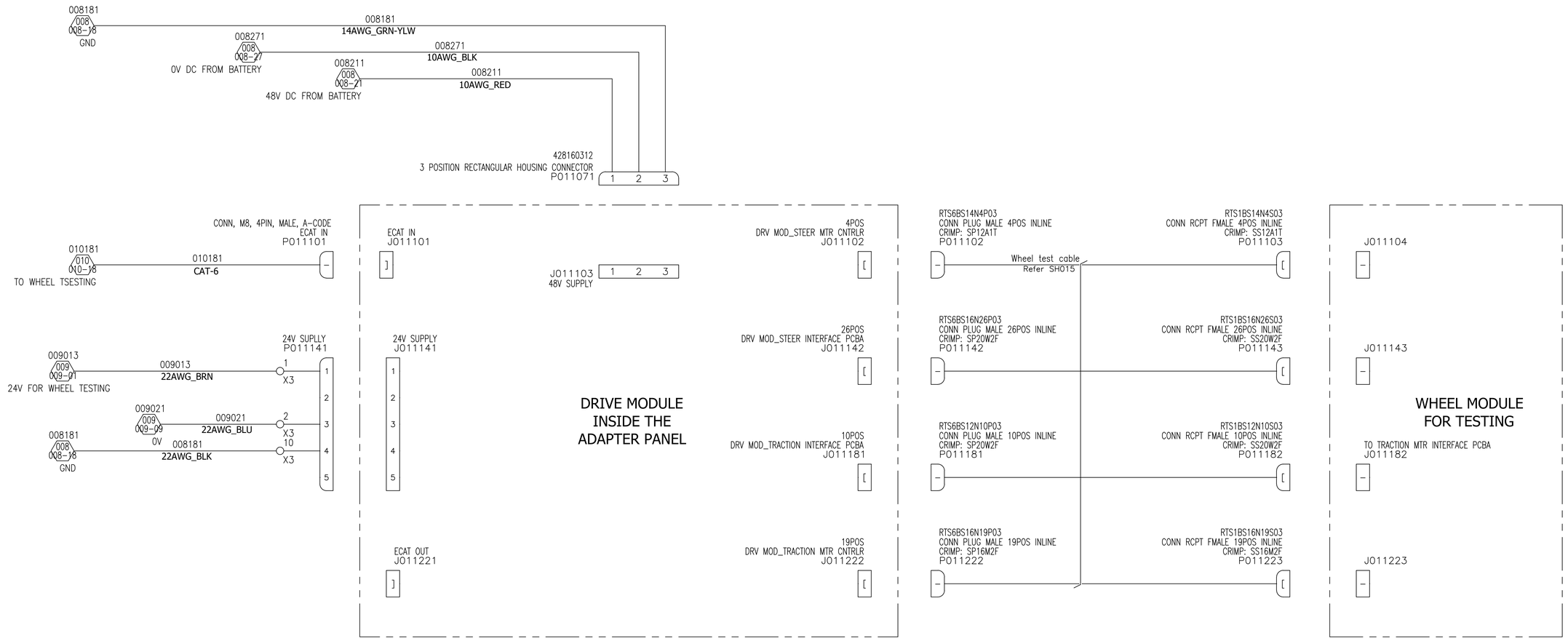




REVISION HISTORY							
REV	DESCRIPTION	DATE	APPROVED				
REV.A	RELEASED FOR PRODUCTION	20/10/2026		DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
REV.0	INITIAL RELEASE	10/10/2026			APPROVED DATE 20/10/2026	SCALE NTC	SHEET 010 OF 015	

DRIVE TRAIN ADAPTER PANEL

IPC TERMINAL CONTROL



GENERAL NOTES:

1.DRIVE MODULE INSTALLATION REQUIREMENT:
- MOUNT THE DRIVE MODULE ON THE RIGHT SIDE OF THE ADAPTER PANEL USING THE PROVIDED BRACKET.

2.FIELD WIRING & INTERCONNECTION INSTRUCTIONS:
- CONNECT 48V DC SUPPLY
- CONNECT 24V DC SUPPLY
- CONNECT ECAT IN

3.WHEEL TESTING PROCEDURE:
- PERFORM WHEEL TESTING AS PER THE INSTRUCTIONS VALIDATED ON THE HMI.
- CONNECT THE WHEEL TESTING HARNESS DIRECTLY TO THE TESTING WHEEL.
- AFTER OPERATION, REMOVE THE CABLE AND HANG IT ON THE PROVIDED J-HOOK.

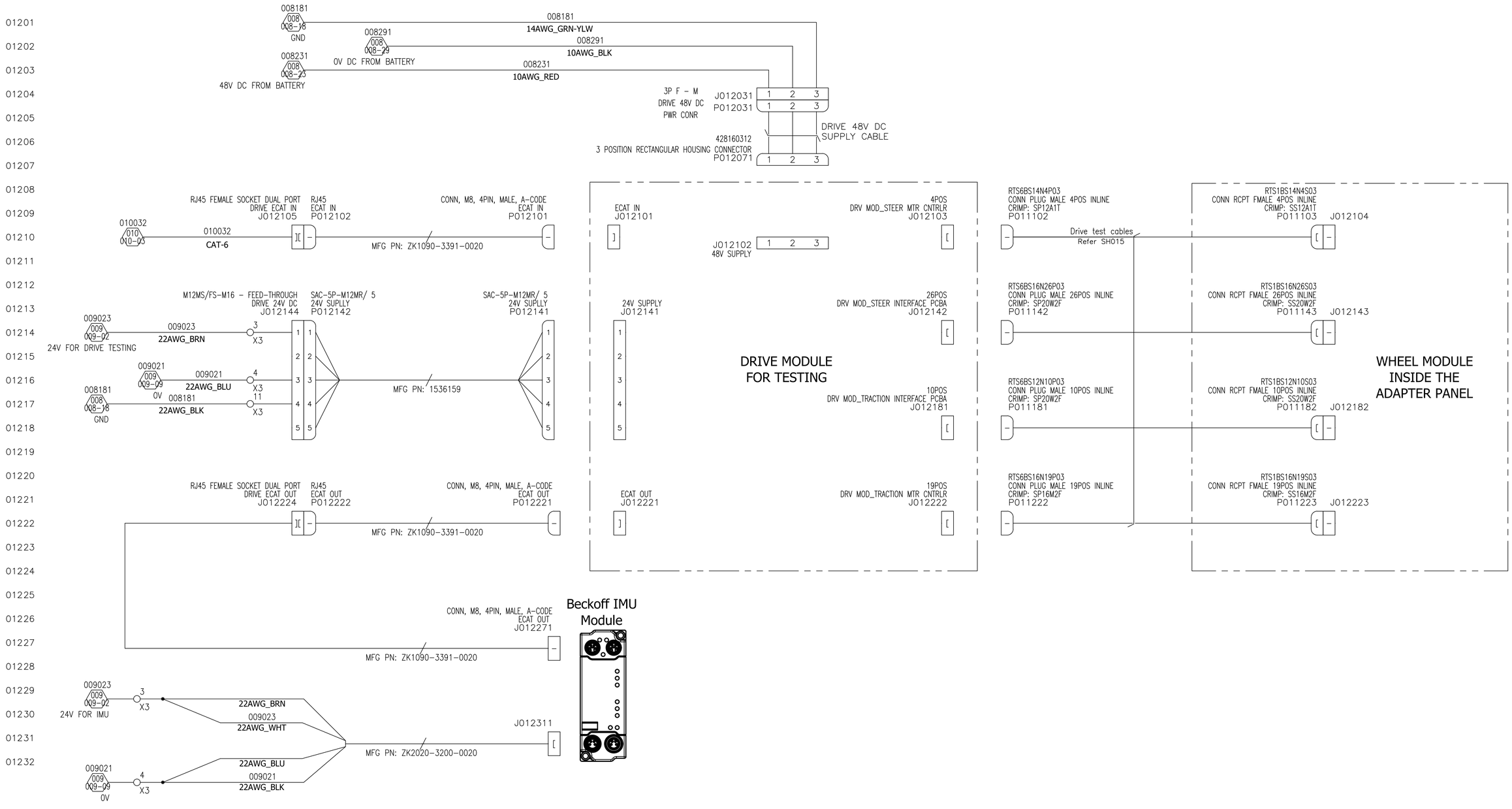
REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

FIELD CONNECTIONS

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 011 OF 015	



GENERAL NOTES:

1.DRIVE MODULE TEST & COMPONENT MOUNTING:

- MOUNT THE WHEEL MODULE INSIDE THE ADAPTER PANEL.
- CONNECT THE DRIVE TEST CABLES TO THE WHEEL MODULE.
- MOUNT THE IMU MODULE (BECKHOFF IMU MODULE) INSIDE THE PANEL.
- CONNECT THE POWER SUPPLY AND ECAT OUT CONNECTORS PROVIDED IN THE PANEL TO IMU MODULE.

2.DRIVE MODULE TESTING PROCEDURE:

- PLACE THE DRIVE MODULE FOR TESTING IN THE RIGHT SIDE PANEL HOLDER.
- FOLLOW ALL PROMPTS AND INSTRUCTIONS DISPLAYED ON THE HMI SYSTEM TO COMPLETE THE DRIVE MODULE TESTING CYCLE.
- AFTER THE TEST COMPLETION, REMOVE THE 48V CABLES HARNESS AND SECURE IT SAFELY IN THE PROVIDED J-HOOK.

REVISION HISTORY

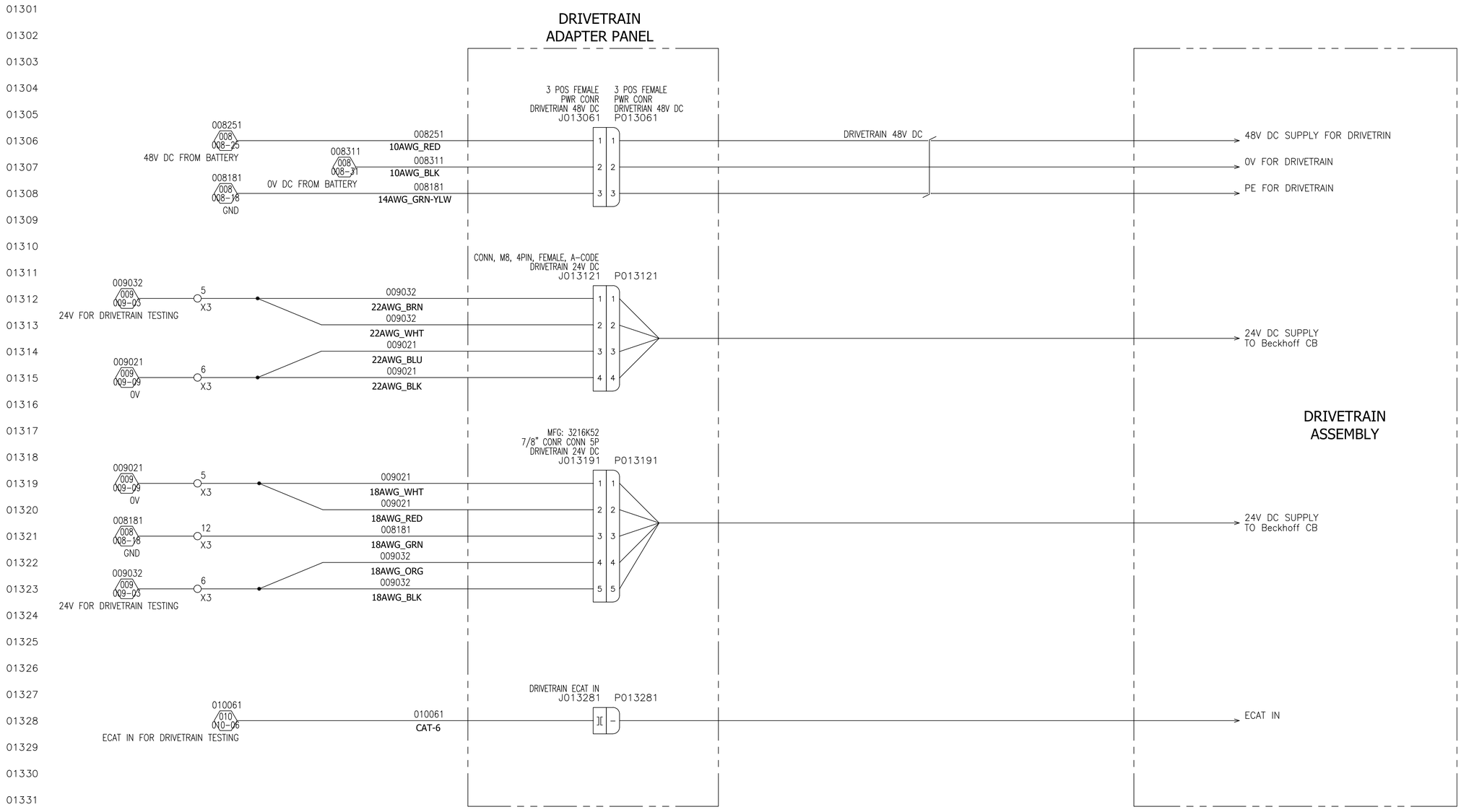
REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

FIELD CONNECTIONS

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER
APPROVED	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 012 OF 015

REV A



GENERAL NOTES:
1.DRIVE TRAIN TESTING PROCEDURE:
- FOLLOW ALL PROMPTS AND INSTRUCTIONS DISPLAYED ON THE HMI SYSTEM TO COMPLETE THE DRIVE TRAIN TESTING CYCLE.
- AFTER THE TEST COMPLETION, REMOVE THE 48V CABLES HARNESS AND SECURE IT SAFELY IN THE PROVIDED J-HOOK.

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

FIELD CONNECTIONS

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 013 OF 015

REV A

SL No.	Description	Make	PartNumber	QTY
1	EtherCAT Coupler	BECKHOFF	EK1101-0010	1
2	EtherCAT Terminal, 4-channel digital input, 24 V DC, TwinSAFE	BECKHOFF	EL1904	1
3	16-Channel Digital Output Terminal, 24 V DC	BECKHOFF	EL2409	1
4	EtherCAT Terminal communication interface, TwinSAFE Logic	BECKHOFF	EL6910	1
5	EtherCAT Junction / 2-Port EtherCAT Coupler	BECKHOFF	EK1122	2
6	EtherCAT Bus End Cover / End Terminal	BECKHOFF	EL9011	1
7	EtherCAT Box, 1-channel, 1 x 3-axis accelerometer, 1 x 3-axis gyroscope	BECKHOFF	EP3751-0260	1
8	M8 4P male right-angle → RJ45 8P male straight, PUR AWG26, drag-chain suitable	BECKHOFF	ZK1090-3391-0010	2
9	M8 4P male right-angle → RJ45 8P male straight, PUR AWG26, drag-chain suitable	BECKHOFF	ZK1090-3391-0020	2
10	M8 4P male right-angle → RJ45 8P male straight, PUR AWG26, drag-chain suitable	BECKHOFF	ZK1090-3391-0030	2
11	EtherCAT cable, PUR, AWG26, drag-chain suitable, M8 4P male straight → RJ45 male straight, A-coded 1 m	BECKHOFF	ZK1090-3191-2010	3
12	EtherCAT cable, PUR, AWG26, drag-chain suitable, M8 4P male straight → RJ45 male straight, A-coded 3 m	BECKHOFF	ZK1090-3191-2030	2
13	Power cable, PUR, 4 x 0.34 mm², drag-chain suitable	BECKHOFF	ZK2020-3200-0020	1
14	PT 6-QUATTRO-PE - Protective conductor terminal block	Phoenix	3212950	50 Piece
15	ZB 8,LGS:L1-N,PE - Marker for terminal blocks	Phoenix	1052413	20 Piece
16	PT 4-QUATTRO - Feed-through terminal block	Phoenix	3211797	50 Piece
17	FBS 2-6 - Plug-in bridge	Phoenix	3030336	50 Piece
18	D-PT 4-QUATTRO - End cover	Phoenix	3208979	50 Piece
19	ZB 6,LGS:FORTL.ZAHLEN marker strip	Phoenix	1051016	1
20	PT 4-QUATTRO-PE - Protective conductor terminal block	Phoenix	3211809	50 Piece
21	PT 1,5/S-QUATTRO - Feed-through terminal block	Phoenix	3208197	50 Piece
22	FBS 2-3,5 - Plug-in bridge	Phoenix	3213014	50 Piece
23	D-PT 1,5/S-QUATTRO - End cover	Phoenix	3208375	50 Piece
24	ZB 3,5,LGS:FORTL.ZAHLEN - Zack marker strip	Phoenix	801404	1
25	PLC-RPT- 24DC/21 - Relay module	Phoenix	2900299	20 Piece
26	FBST 50-PLC RD - Single plug-in bridge	Phoenix	1081050	20 Piece
27	CBMC E4 24DC/1-4A NO - Electronic circuit breaker	Phoenix	2906031	1
28	SACC-5P-DSI-M12MS/FS-M16 - Control cabinet feed-through	Phoenix	1551671	6
29	SAC-5P-M12MR/ 5,0-PUR SH - Sensor/actuator cable	Phoenix	1682773	1
30	SAC-5P-M12MR-M12FR SH/.../... - Sensor/actuator cable	Phoenix	1536159	1
31	SAC-5P- 1,5-PUR/M12FR SH - Sensor/actuator cable	Phoenix	1682964	6
32	SAC-5P-M12MS/ 0,6-PUR/M12FS SH - Sensor/actuator cable	Phoenix	1500897	4
33	SACC-M12FR-4CON-PG7-M - Connector	Phoenix	1681143	1
34	3 Position Rectangular Housing Connector Receptacle Black 0.394" (10.00mm)	Molex	428160312	3
35	Socket Contact 10-12 AWG Crimp	Molex	428150012	12
36	HC-STA-D07-BWSCF-PL-BK - Housing	Phoenix	1419277	3
37	HC-Q02-I-AT-F - Contact insert	Phoenix	1496732	3
38	HC-STA-D07-HHFS-1TGM25-PL-BK - Housing	Phoenix	1419256	3
39	HC-Q02-I-AT-M - Contact insert	Phoenix	1496723	3
40	Battery connector	--	--	2
41	19 Position	Amphenol	RTS6BS16N19P03	5
42	19 Position	Amphenol	RTS1BS16N19S03	5
43	10 Position	Amphenol	RTS6BS12N10P03	5
44	10 Position	Amphenol	RTS1BS12N10S03	5
45	26 Position	Amphenol	RTS6BS16N26P03	5
46	26 Position	Amphenol	RTS1BS16N26S03	5
47	4 Position	Amphenol	RTS6BS14N4P03	5
48	4 Position	Amphenol	RTS1BS14N4S03	5
49	Crimp Contacts	Amphenol	SP16M2F	125
50	Crimp Contacts	Amphenol	SS16M2F	125

SL No.	Description	Make	PartNumber	QTY
51	Crimp Contacts	Amphenol	SP20W2F	65
52	Crimp Contacts	Amphenol	SS20W2F	65
53	Crimp Contacts	Amphenol	SP20W2F	155
54	Crimp Contacts	Amphenol	SS20W2F	155
55	Crimp Contacts	Amphenol	SP12A1T	28
56	Crimp Contacts	Amphenol	SS12A1T	28
57	Drive Module Fully Assembled GEN2	--	--	1
58	Fuse Holder HOLDER MEGA FUSE WITH FLEXIBLE COVER	Littelfuse	02981028HXFC	2
59	Automotive Fuses Fuse - Mega 70V (SF51) 2-M8 holes 60A	Littelfuse	0998060.UX-2M8	2
60	Automotive Fuses MEGA 70V (SF51)2-M8 HOLES 100A	Littelfuse	0998100.UX-2M8	1
61	Fuse Block 30 A 80V 12 Circuit Blade Chassis Mount	Optifuse	BLR-1212-G	1
62	Contactors - Electromechanical CONTACTOR	Sensata	GX14CB	1
63	Schottky Diodes & Rectifiers 200V 250A Module SOT-227	Vishay Semiconductors	VS-QA250FA20	1
64	30 A 80 V Fuse Automotive Requires Holder Blade, ATO/ATC	OptiFuse	ANR80-UL-30A	5
65	Power Supplies - Chassis Mount 600W 48V 12.5A	Cosel	Cosel PJA600F-48	1
66	Yellow/Green Flexible Single-Core Copper Wire	Polycab/ Iapp	6 mm²	30 Mtr
67	Red Flexible Single-Core Copper Wire	Polycab/ Iapp	6 mm²	30 Mtr
68	Black Flexible Single-Core Copper Wire	Polycab/ Iapp	6 mm²	30 Mtr
69	Blue Flexible Single-Core Copper Wire	Polycab/ Iapp	6 mm²	30 Mtr
70	Red Flexible Single-Core Copper Wire	Polycab/ Iapp	10 mm²	15 Mtr
71	Black Flexible Single-Core Copper Wire	Polycab/ Iapp	10 mm²	15 Mtr
72	6 SQMM Wire Ferrules Copper Crimp Connectors, Pack of 100 Pcs	--	--	100 Piece
73	6 mm² × M5 ring lug	--	--	20 Piece
74	10 mm² × M5 ring lug	--	--	20 Piece
75	10 mm² × M8 ring lug	--	--	20 Piece
76	Transparent Heat Shrink Sleeve 15 mm	--	--	10 Mtr
77	Transparent Heat Shrink Sleeve 25 mm	--	--	5 Mtr
78	Transparent Heat Shrink Sleeve 35 mm	--	--	5 Mtr
79	Monolithic pilot light, Harmony XB5N XB7N, plastic, green, 22mm, LED, 230V AC	Schneider Electric	XB7EV03MPN	3
80	Exhaust Fan 230VAC	Rexnord	REC-22038 A2T	2
81	4" Louvers, Screw fit (130 x 130mm)	Elettro	ETAV-01	4
82	Fan Grill (130 x 130mm)	--	--	2
83	YT-RJ45-JSX-16-001-CNLINKO-YT RJ45 Female Socket Dual Port Metal Panel Ethernet Connector IP67 Panel Mount	CNLINKO	---	10
84	Cable duct 25x80	---	---	10 Mtr
85	Cable duct 25x25	---	---	2 Mtr
86	Cable duct 25x40	---	---	4 Mtr
87	Cable duct 25x60	---	---	4 Mtr
88	DIN rail	---	---	5 Mtr
89	AC/DC Single output Industrial DIN rail power supply - Output 24Vdc at 20A - 480W - metal case.	Mean Well	NDR-480-24	2
90	Modular Cable Plug to Plug, Angled Right, 90° 8p8c (RJ45, Ethernet) 5.00' (1.52m) Unshielded	Eaton Tripp Lite	N204-S05-BL-RA	10
91	Modular Cable Plug to Plug, Angled Right, 90° 8p8c (RJ45, Ethernet) 3.00' (914.4mm) Unshielded	Eaton Tripp Lite	N204-S03-BL-RA	1
92	Test Clips Plain Grounding Steel Clip - 100 Amp	Mueller Electric	BU-11APN	1
93	Test Clips Green Insulator for BU-11 Clip	Mueller Electric	BU-13-5	1
94	6 sq mm 4 Core Copper Flexible Wire	---	---	2 Mtr
95	RIGHT-ANGLE CAT6 GIGABIT MOLDED	Eaton Tripp Lite	N204-S02-BL-RA	10
96	SACC-DSI-M8FS-4CON-M10/0,5 - Device connector rear mounting	Phoenix	1456093	1
97	Mini Connector Female Panel-Mount Receptacle, 5 Poles, 1' Long Wire Leads	McMaster-Carr	3216K52	1

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRAWN VIJAYAKUMAR V S

DRAWN DATE 10/10/2026

APPROVED DATE 20/10/2026

DRIVE TRAIN ADAPTER PANEL

BILL OF MATERIALS (BOM)

SIZE D | DRAWING NUMBER | REV A

SCALE NTC | SHEET 014 OF 015

01501
01502
01503
01504
01505
01506
01507
01508
01509
01510
01511
01512
01513
01514
01515
01516
01517
01518
01519
01520
01521
01522
01523
01524
01525
01526
01527
01528
01529
01530
01531
01532

RTS6BS16N19P03
19 Position Circular Connector Plug Housing
SP16M2F CONTACT PIN 16-18AWG SZ16 CRIMP

19 core 16-18 AWG (1 sqmm)
pin to pin connection

RTS1BS16N19S03
CONN RCPT HSG FMALE 19POS INLINE
SS16M2F Socket Contact 16-18 AWG Size 16 Crimp Gold

RTS6BS12N10P03
CONN PLUG HSG MALE 10POS INLINE
SP20W2F CONTACT PIN 20-22AWG SZ20 CRIMP

10 core 20-22AWG (0.5 sqmm)
pin to pin connection

RTS1BS12N10S03
CONN RCPT HSG FMALE 10POS INLINE
SS20W2F CONTACT SOCKET 20-22AWG SZ20

RTS6BS16N26P03
CONN PLUG HSNG MALE 26POS INLINE
SP20W2F CONTACT PIN 20-22AWG SZ20 CRIMP

19 core 16-18 AWG (1 sqmm)
pin to pin connection

RTS1BS16N26S03
CONN RCPT HSG FMALE 26POS INLINE
SS20W2F CONTACT SOCKET 20-22AWG SZ20

RTS6BS14N4P03
CONN PLUG HSNG MALE 4POS INLINE
SP12A1T CONN PIN 12-14AWG TIN CRIMP

19 core 16-18 AWG (1 sqmm)
pin to pin connection

RTS1BS14N4S03
CONN RCPT HSG FMALE 4POS INLINE
SS12A1T CONN SOCKET 12-14AWG TIN CRIMP

TECHNICAL NOTE:
This batch consists of 4 distinct cable configurations based on pin counts (19-Pin, 10-Pin, 8-Pin, and 4-Pin) using Amphenol EcoMate Aquarius connectors. Each of the 4 configurations requires 2 identical sets (1 set for Drive Testing, 1 set for Wheel Testing). All finished assemblies must strictly follow a 2-Meter length constraint with pin-to-pin wiring configuration.

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
REV.A	RELEASED FOR PRODUCTION	20/10/2026	
REV.0	INITIAL RELEASE	10/10/2026	

DRIVE TRAIN ADAPTER PANEL

CABLE HARNESS

DRAWN VIJAYAKUMAR V S	DRAWN DATE 10/10/2026	SIZE D	DRAWING NUMBER	REV A
	APPROVED DATE 20/10/2026	SCALE NTC	SHEET 015 OF 015	